ALGEBRA 1 - UNIT 2

Section 2.2
Graphing Linear Functions

Turning a rule into a picture of constant change.

> **Mathematical idea:** A line tells a story: where the relationship starts, how it changes, and what values are possible.

Today we will build linear graphs from tables, equations, and situations. We will focus on what the graph means, not just drawing a line.

Learning Target

Students will graph linear functions and interpret key features.

I can statements

- I can graph linear functions.
- I can identify slope and intercepts from a graph.
- I can determine whether a line is increasing or decreasing.
- I can connect graphs to situations.
- I can use slope and intercepts to build a graph.

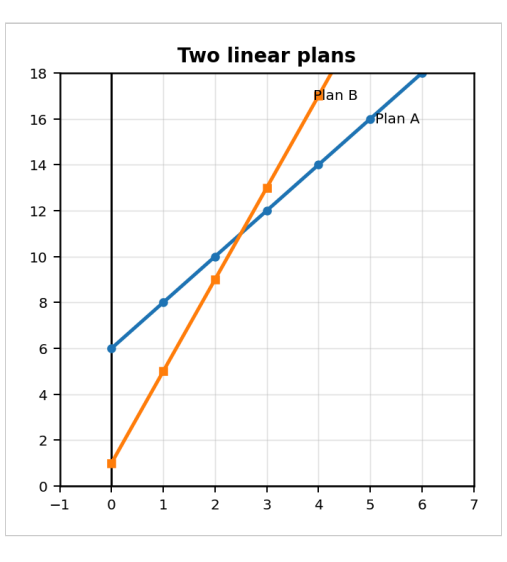
What's To Come

This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: Where does each graph start? Which graph changes faster? Does the lower starting value always stay lower? What graph evidence could support a recommendation?

Future Task

Two streaming plans are shown on the graph. A student says, “Plan B is always cheaper because it starts lower.”



1. Use graph evidence to decide whether the student’s claim is always true.
2. Explain why starting value and rate of change both matter.
3. Write a recommendation that depends on how long someone plans to use the plan.
4. Explain what information from an equation, table, or context would make your recommendation stronger.

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

A linear function makes a straight-line graph. The line is straight because the relationship changes at a constant rate. That means the output changes by the same amount again and again when the input changes by equal steps.

A graph can show important features quickly. The **intercept** shows where the line crosses an axis. The y-intercept often shows the starting value when $x = 0$. The **slope** shows how much the output changes for each 1-unit change in the input.

There is more than one way to graph a linear function. You can make a table of input-output pairs. You can start at the y-intercept and use the slope. You can also use a context to decide what the starting value and rate of change mean.

A good graph is not just a drawing. It is a model. The axes, scale, points, and line should match the situation and help someone understand the relationship.

Stop and Discuss

1. Why does a constant rate of change make a straight-line graph?
2. What information can the y-intercept give us?
3. Why should a graph’s scale and labels match the situation?

A Table Gives Points for the Graph

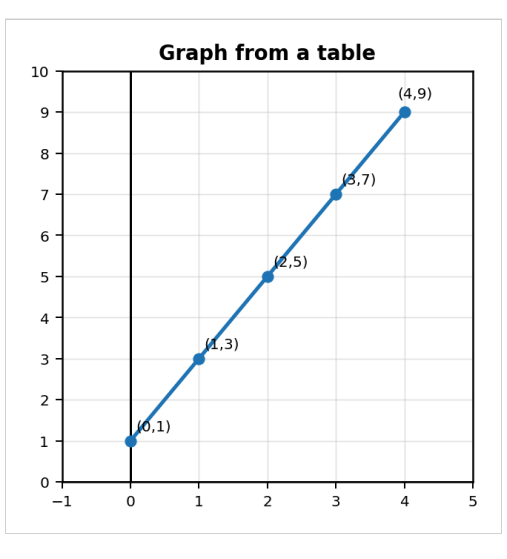
A table lists input-output pairs. Each pair becomes a point on the coordinate plane. After plotting the points, check whether they form a straight line.

Linear graph: A graph that forms a straight line because the rate of change is constant.

Example 1

Use the table to graph the linear function.

x	0	1	2	3	4
y	1	3	5	7	9



Explain how the table and graph both show the same starting value and constant change.

You Try It 1

Complete the table for $y = 3x + 2$ using $x = 0, 1, 2,$ and 3. Then describe what the graph should look like before you sketch it.

x	0	1	2	3
y				

Stop and Discuss

1. How does each table column become a point on the graph?
2. Why is it helpful to predict the shape before graphing?

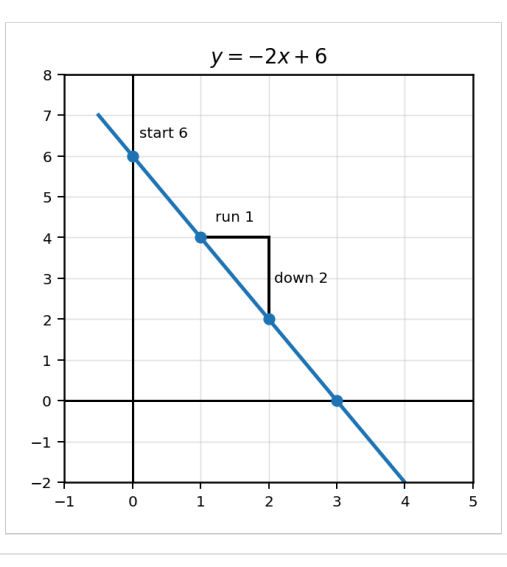
Slope and Intercept Can Build the Line

In slope-intercept form, $y = mx + b$, the value of b is the y-intercept and the value of m is the slope. Start at the intercept, then use the slope to find more points.

Graphing idea: Start at b on the y-axis. Then use m as rise over run to move to another point.

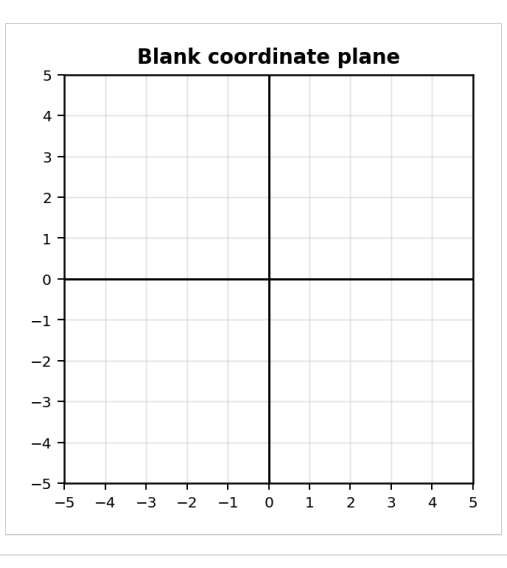
Example 2

Graph $y = -2x + 6$ using the y-intercept and slope. Then explain what the slope tells the graph to do.



You Try It 2

Graph $y = \frac{1}{2}x + 3$ on the blank grid. Mark the y-intercept first, then use the slope to find at least two more points.

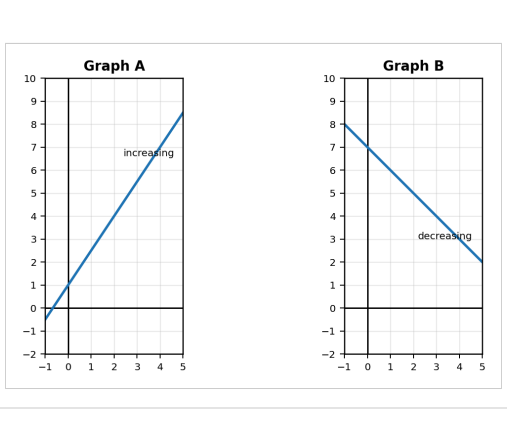


Graphs Show Direction and Key Features

A line can increase, decrease, or stay flat. The direction depends on the slope. The intercepts and points help us describe what the graph means.

Example 3

Compare the two graphs. Identify which line is increasing and which line is decreasing. Then explain how the slope affects the direction.



You Try It 3

A line has a positive slope and crosses the y-axis at -4. Describe what the graph should do. Then sketch a possible graph and label the intercept.

Stop and Discuss

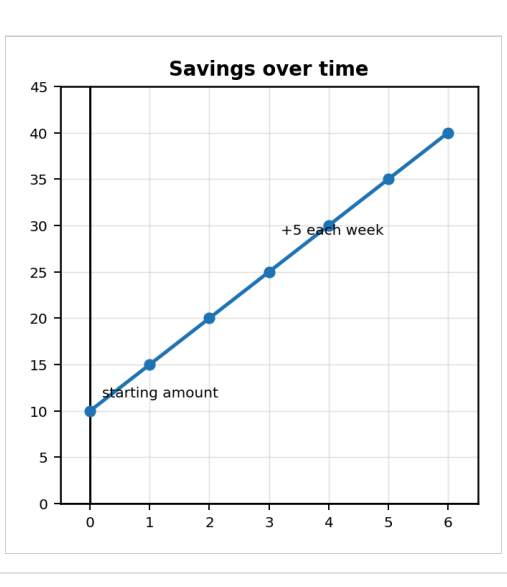
1. How can you tell the sign of the slope by looking at the graph?
2. Why can two lines have the same intercept but different slopes?

A Graph Should Match the Situation

When a graph represents a context, the axes, starting value, and rate of change should all make sense. A graph can help us make predictions, but the prediction should be reasonable for the situation.

Example 4

A student starts with $\backslash 10insavingsandadds$5 each week. Use the graph to explain the starting value, the rate of change, and one reasonable prediction.



You Try It 4

A plant is 6 cm tall and grows 2 cm each week. Sketch a graph that matches the situation. Label both axes and explain what the y-intercept and slope mean.

Stop and Discuss

1. What graph feature shows the starting savings amount?
2. What graph feature shows the amount added each week?
3. Why might a graph prediction stop being reasonable far outside the situation?

Summary

- A linear function graphs as a straight line.
- A table gives ordered pairs that can be plotted on a graph.
- In $y = mx + b$, b gives the y-intercept and m gives the slope.
- A positive slope makes a line increase from left to right.
- A negative slope makes a line decrease from left to right.
- The y-intercept often represents a starting value in a context.
- A good graph uses labels, scale, and features that match the situation.

Common Mistakes

- **Starting at the slope:** In $y = mx + b$, start at b, not at m.
- **Using slope backward:** Slope is vertical change compared to horizontal change.
- **Ignoring the sign:** A negative slope should move down as you move right.
- **Plotting only one point:** One point is not enough to show a line.
- **Forgetting labels:** In a context, axes should show what the input and output mean.
- **Using a poor scale:** A graph can be hard to read if the scale does not fit the data.

Optional Future Task Reflection

Optional Reflection: Look back at the future task. Do not solve it yet.

1. Highlight one part that makes more sense now than it did at the beginning.
2. Circle one part that still feels unfamiliar.
3. Write one sentence about what representation you would look at first.